

Your first database

INTRODUCTION TO RELATIONAL DATABASES IN SQL



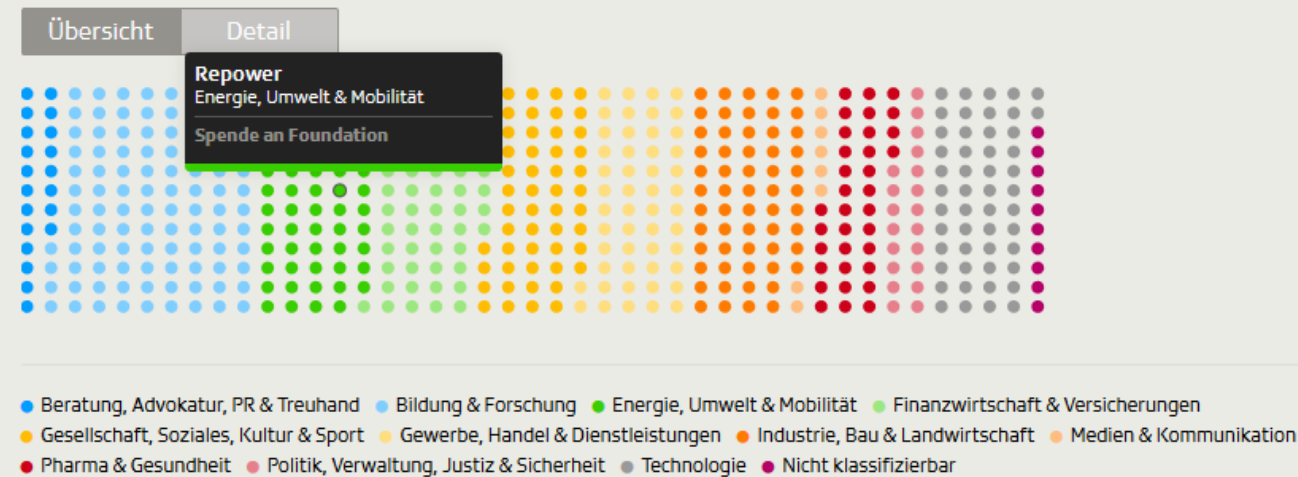
Timo Grossenbacher
Data Journalist

Investigating universities in Switzerland

Eidgenössische Technische Hochschule Zürich

Zu dieser Hochschule gehören rund **18'600 Studierende** und **Professor/innen**. Es besteht ein jährlicher Aufwand von rund **1.6 Mrd. Fr.**, wovon **8.8 %** aus **privaten Drittmitteln** stammen (BFS, 2014).

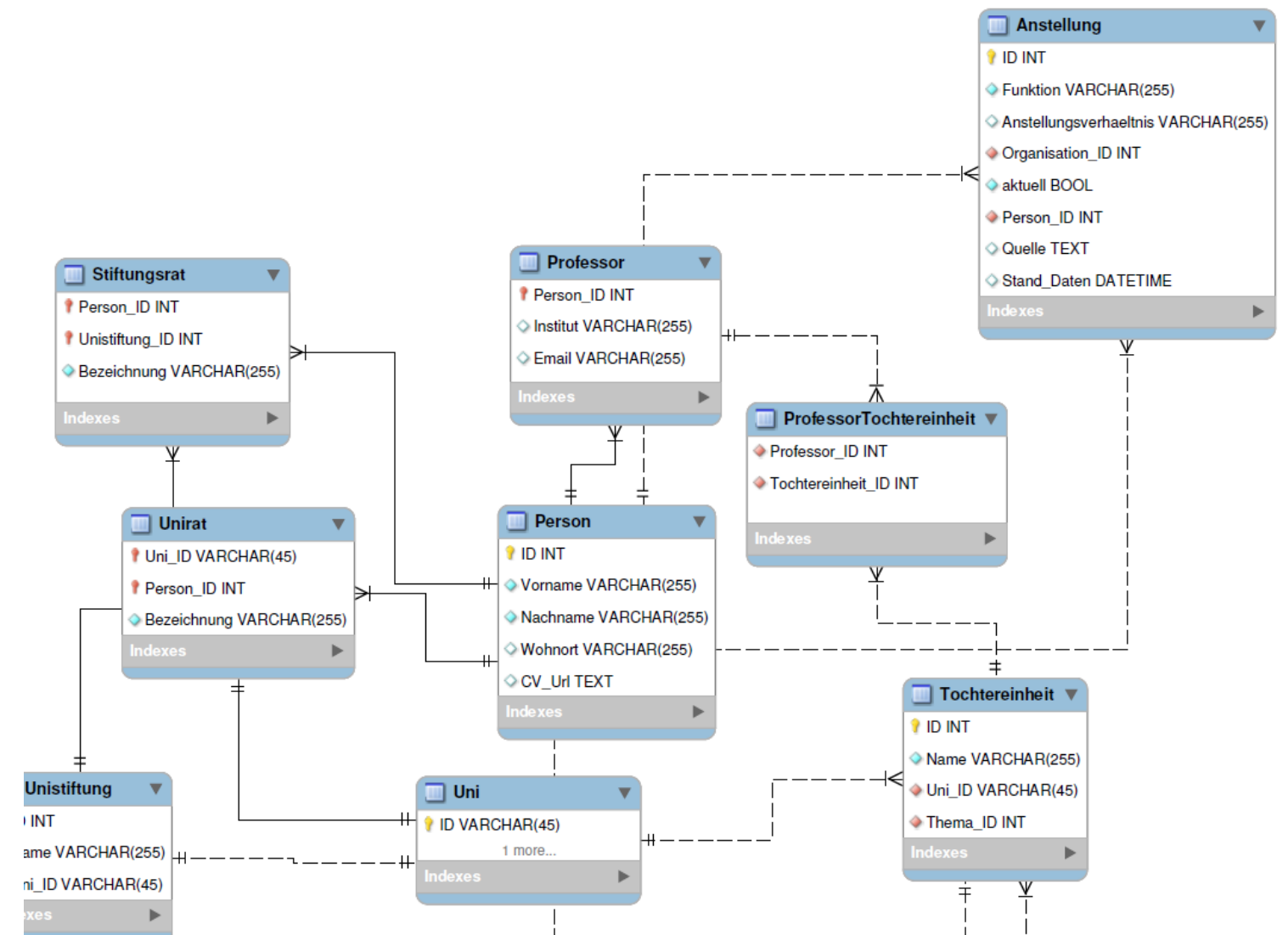
Jeder Punkt in der Grafik zeigt eine von insgesamt **516 Interessenbindungen**.  



Stand der Daten: Juni 2015

Quelle: Pressestelle

Interessenbindungen beinhalten auch Mandate in Verwaltungs- und Stiftungsräten sowie Vereinsvorstände, die im Auftrag der ETH Zürich von Professor/innen wahrgenommen werden. ETHZ und EPFL teilen sich den Universitätsrat («ETH-Rab»).



A relational database:

- real-life *entities* become *tables*
 - reduced redundancy
 - data integrity by *relationships*
- e.g. professors , universities , companies
 - e.g. only one entry in companies for the bank "Credit Suisse"
 - e.g. a professor can work at multiple universities and companies , a company can employ multiple professors

Throughout this course you will:

- work with the data I used for my investigation
- create a relational database from scratch
- learn three concepts:
 - *constraints*
 - *keys*
 - *referential integrity*

You'll need: Basic understanding of SQL, as taught in [Introduction to SQL](#).

Your first duty: Have a look at the PostgreSQL database

```
SELECT table_schema, table_name
FROM information_schema.tables;
```

table_schema	table_name
pg_catalog	pg_statistic
pg_catalog	pg_type
pg_catalog	pg_policy
pg_catalog	pg_authid
pg_catalog	pg_shadow
public	university_professors
pg_catalog	pg_settings
...	

Have a look at the columns of a certain table

```
SELECT table_name, column_name, data_type
FROM information_schema.columns
WHERE table_name = 'pg_config';
```

```
table_name | column_name | data_type
-----+-----+-----
pg_config  | name        | text
pg_config  | setting     | text
```

Let's do this.

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Tables: At the core of every database

INTRODUCTION TO RELATIONAL DATABASES IN SQL

SQL

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Redundancy in the university_professors table

```
SELECT *  
FROM university_professors  
LIMIT 3;
```

```

-[ RECORD 1 ]-----+-----
firstname      | Karl
lastname       | Aberer
university     | ETH Lausanne
university_shortname | EPF
university_city | Lausanne
function       | Chairman of L3S Advisory Board
organization    | L3S Advisory Board
organization_sector | Education & research
-[ RECORD 2 ]-----+-----
firstname      | Karl
lastname       | Aberer
university     | ETH Lausanne
university_shortname | EPF
university_city | Lausanne
function       | Member Conseil of Zeno-Karl Schindler Foundation
organization    | Zeno-Karl Schindler Foundation
organization_sector | Education & research
-[ RECORD 3 ]-----+-----
firstname      | Karl
lastname       | Aberer
(truncated)
function       | Member of Conseil Fondation IDIAP
organization    | Fondation IDIAP
(truncated)

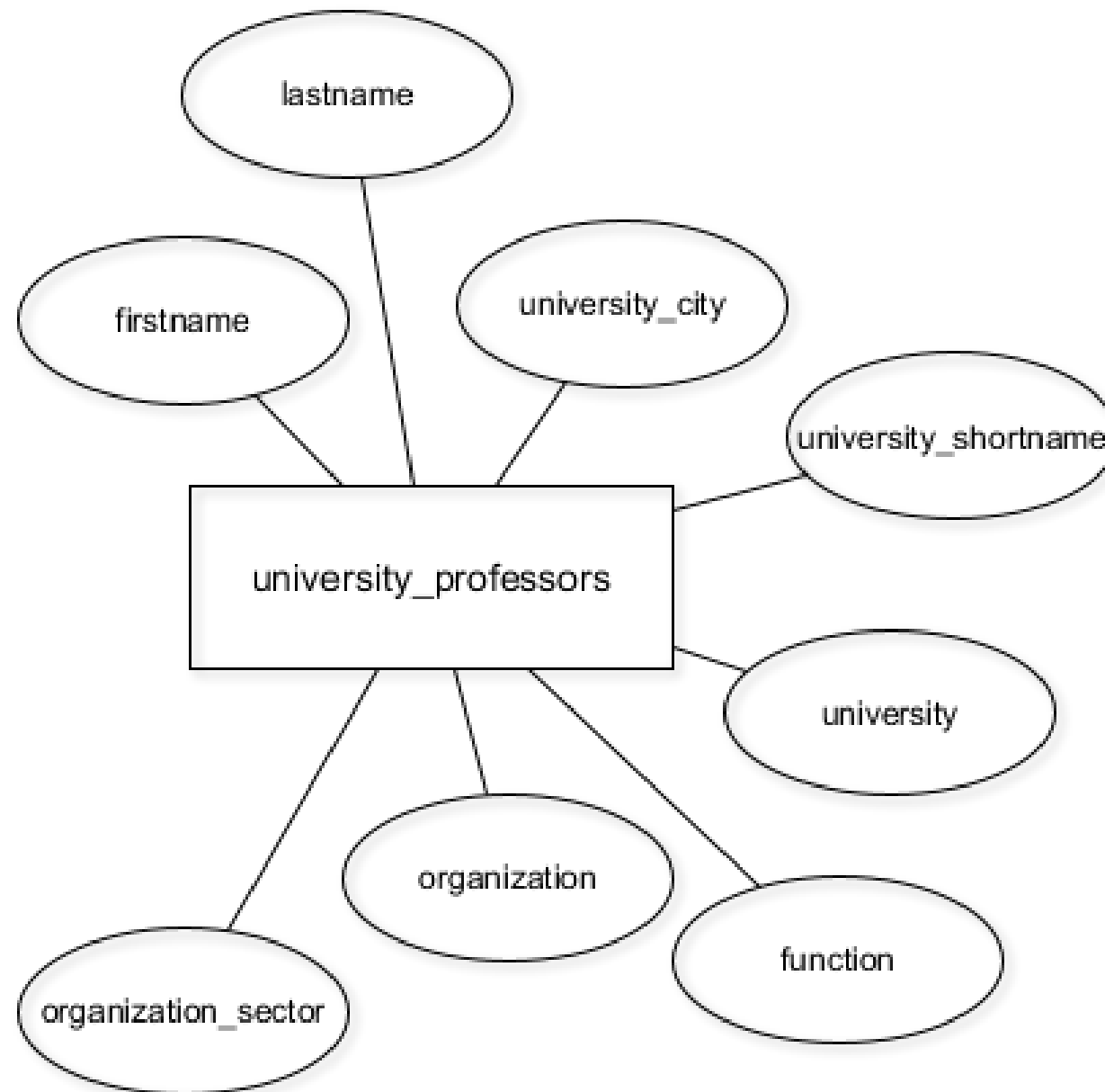
```

```

-[ RECORD 1 ]-----+-----
firstname      | Karl
lastname       | Aberer
university     | ETH Lausanne
university_shortname | EPF
university_city | Lausanne
function       | Chairman of L3S Advisory Board
organisation    | L3S Advisory Board
organisation_sector | Education & research
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organisation    | Zeno-Karl Schindler Foundation
organisation_sector | Education & research
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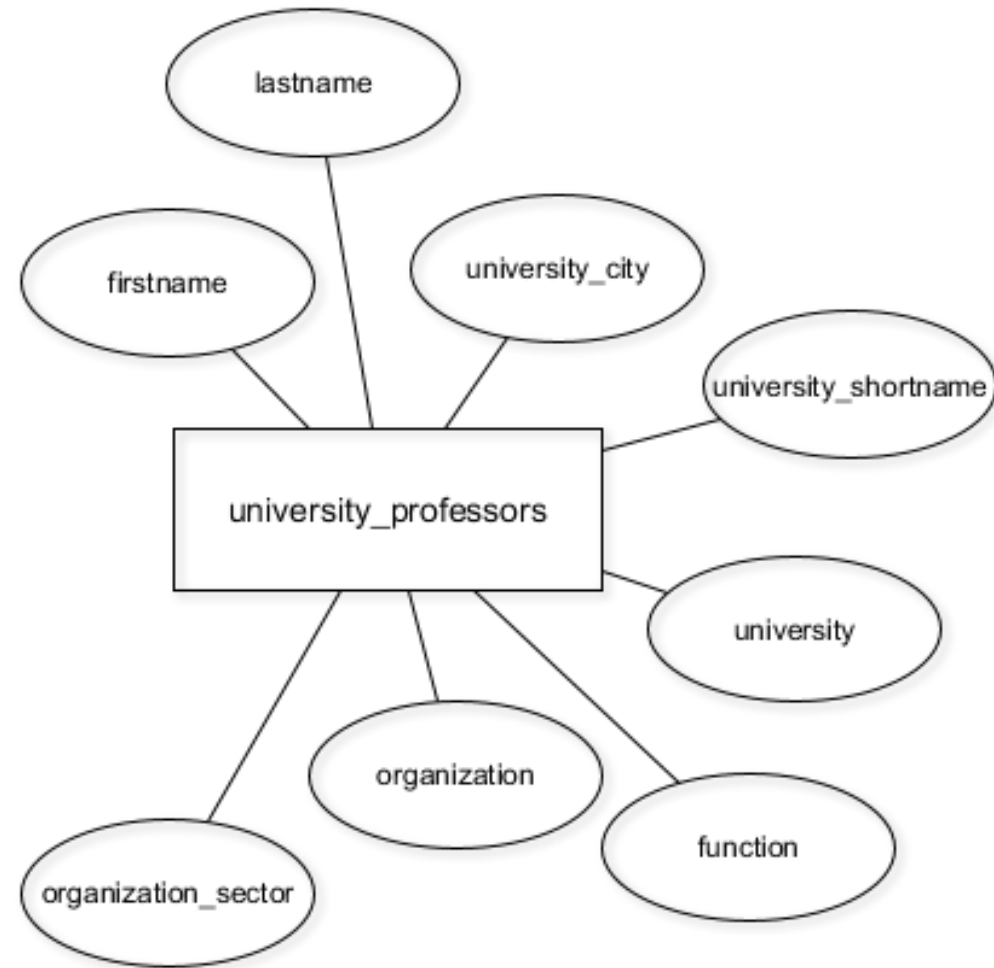
```

Currently: One "entity type" in the database

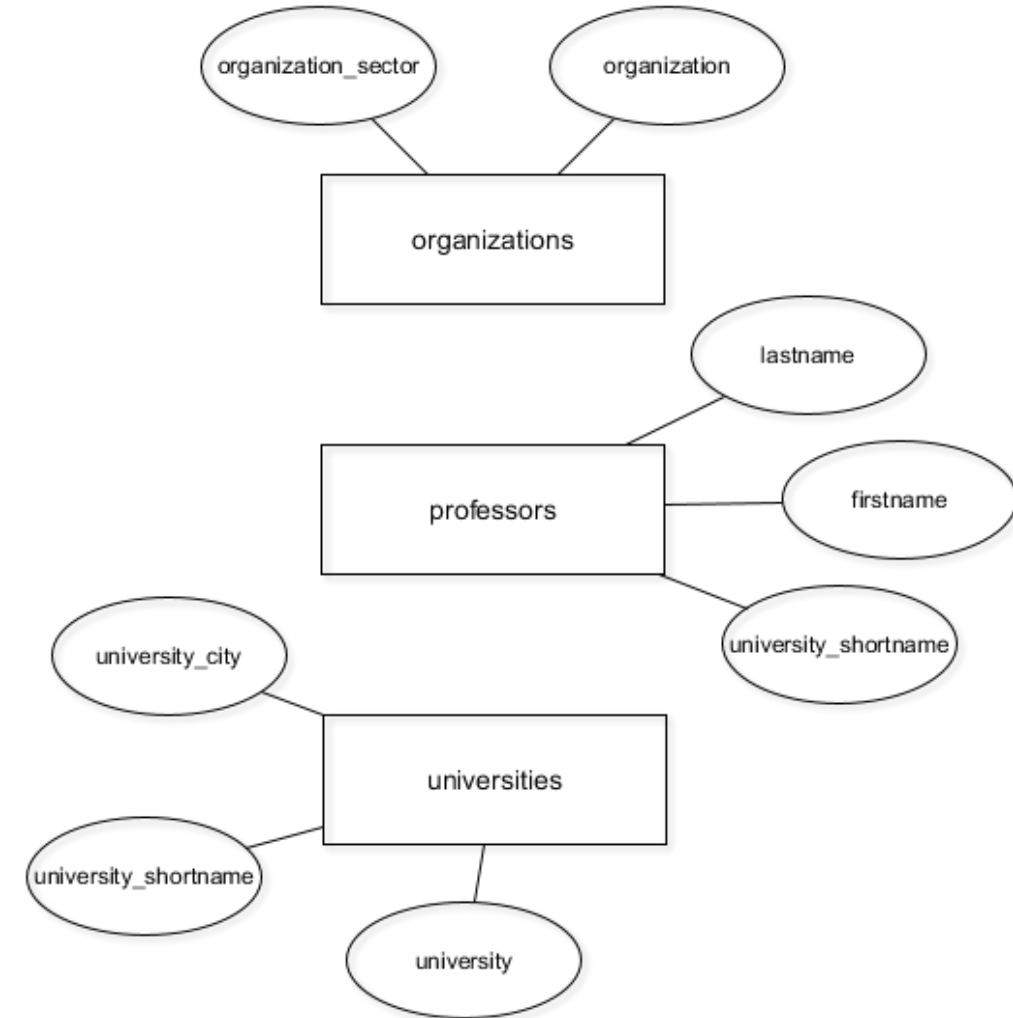


A better database model with three entity types

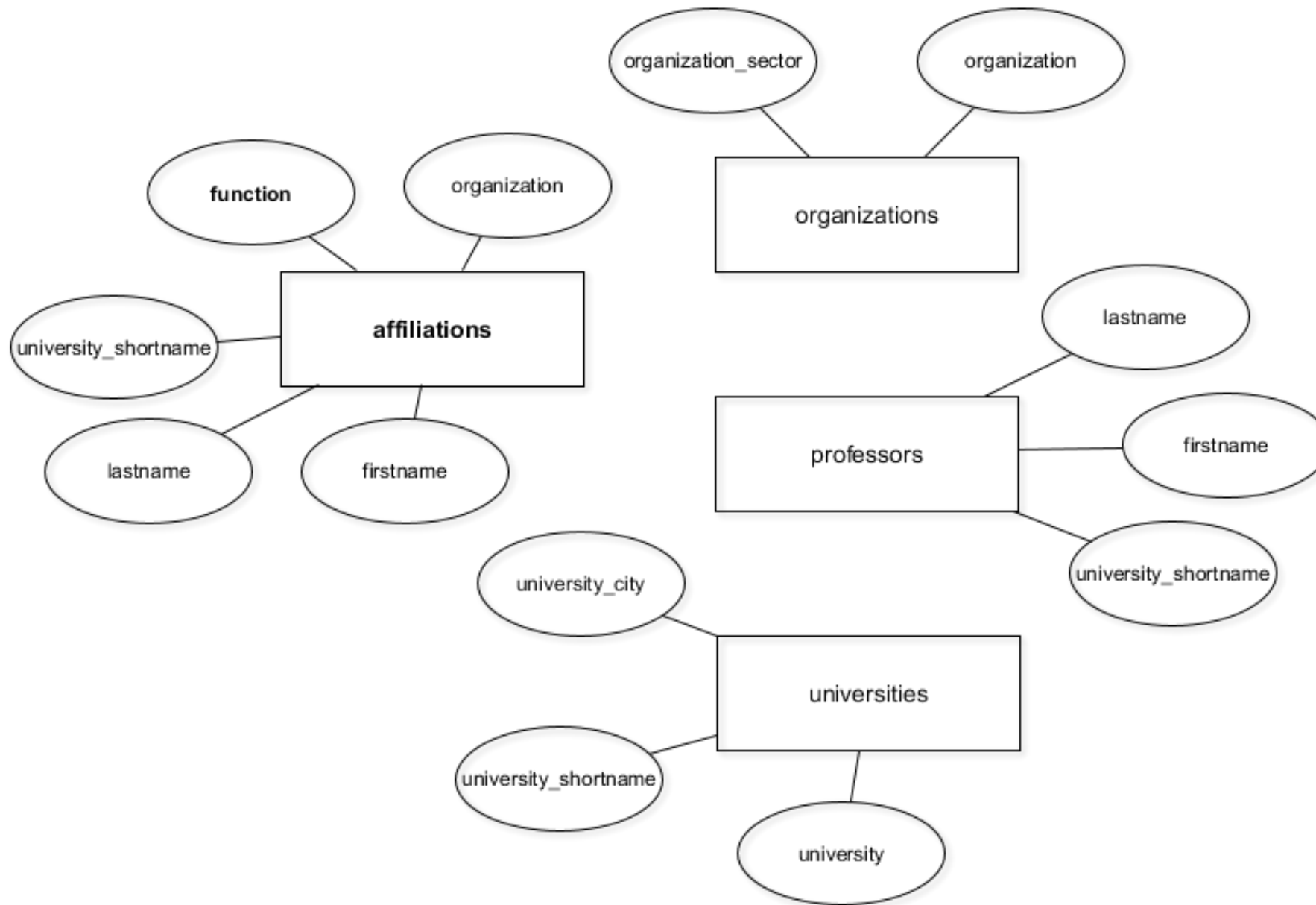
Old:



New:



A better database model with four entity types

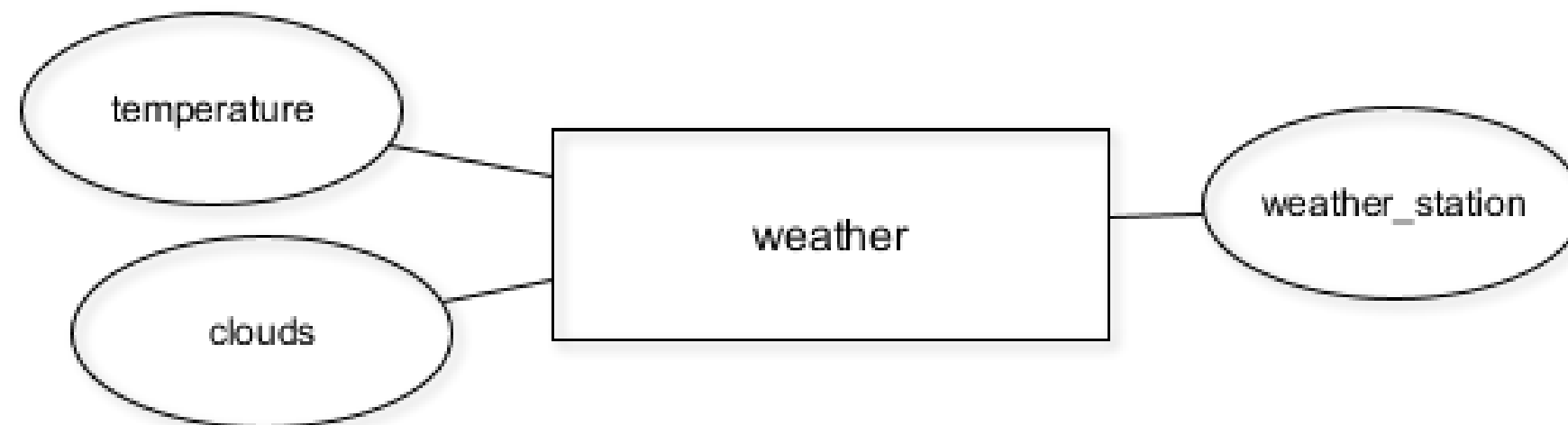


Create new tables with CREATE TABLE

```
CREATE TABLE table_name (  
  column_a data_type,  
  column_b data_type,  
  column_c data_type  
);
```


Create new tables with CREATE TABLE

```
CREATE TABLE weather (  
  clouds text,  
  temperature numeric,  
  weather_station char(5)  
);
```



Let's practice!

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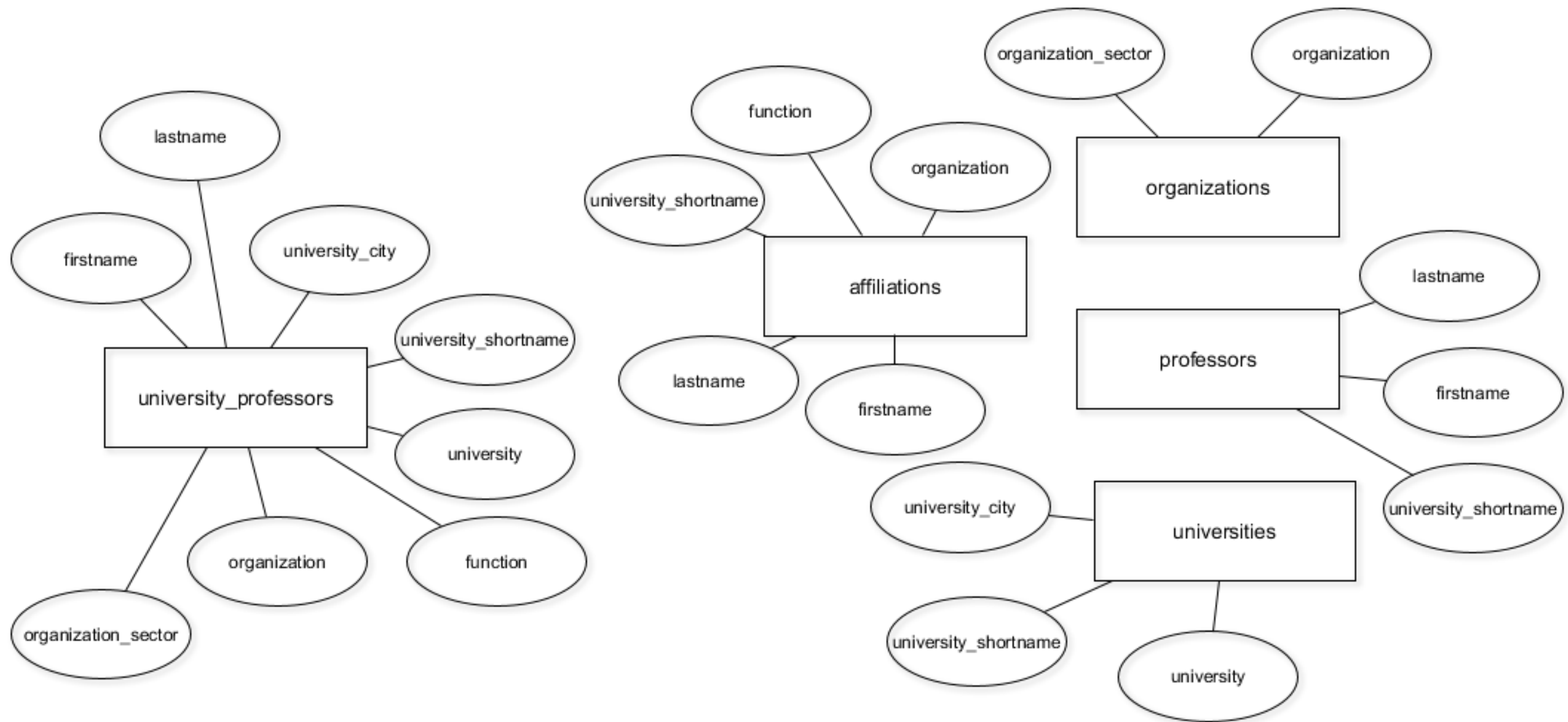
Update your database as the structure changes

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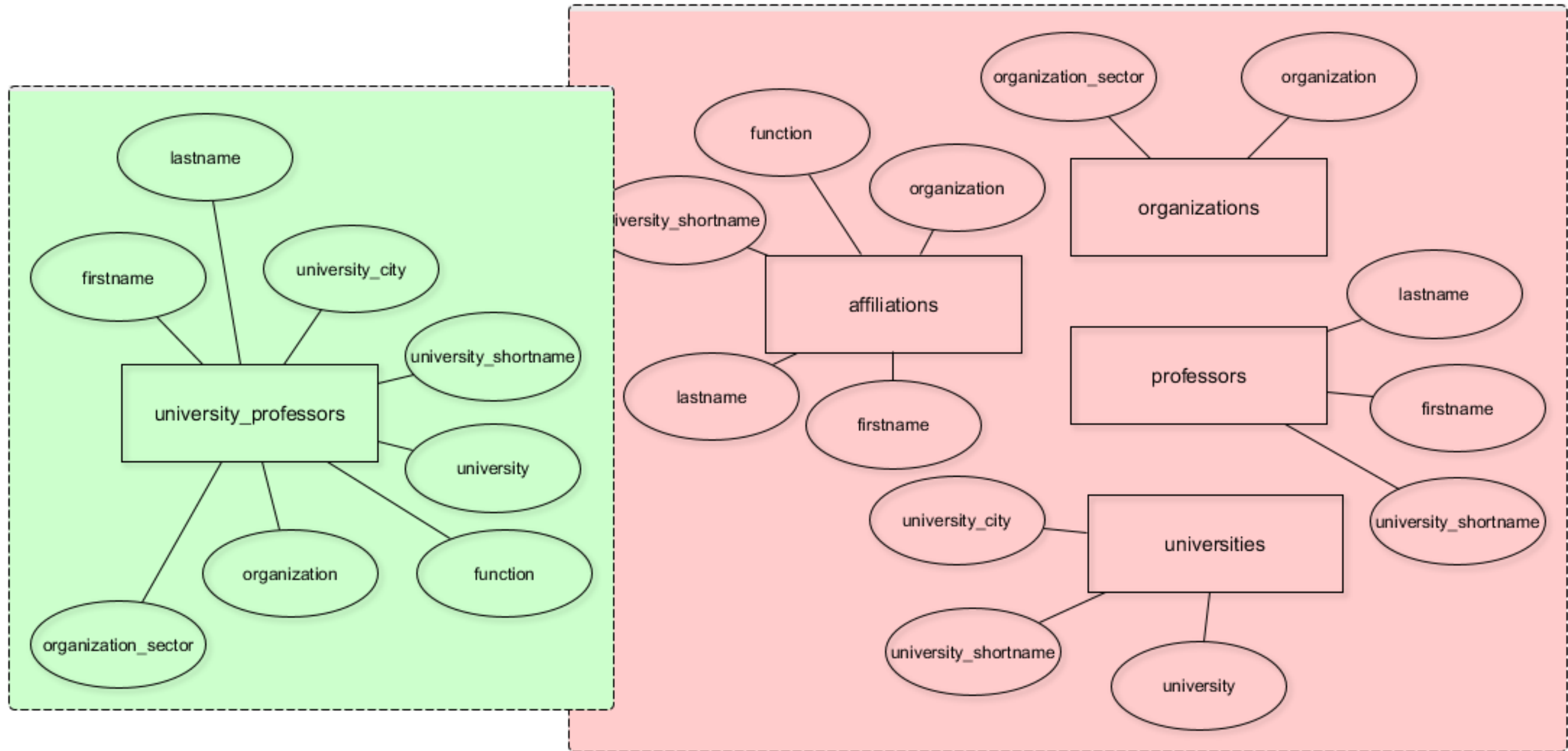
SQL

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The current database model



The current database model



Only store DISTINCT data in the new tables

```
SELECT COUNT(*)  
FROM university_professors;
```

```
count  
-----  
1377
```

```
SELECT COUNT(DISTINCT organization)  
FROM university_professors;
```

```
count  
-----  
1287
```

INSERT DISTINCT records INTO the new tables

```
INSERT INTO organizations
SELECT DISTINCT organization,
               organization_sector
FROM university_professors;
```

Output: INSERT 0 1287

```
INSERT INTO organizations
SELECT organization,
               organization_sector
FROM university_professors;
```

Output: INSERT 0 1377

The INSERT INTO statement

```
INSERT INTO table_name (column_a, column_b)  
VALUES ("value_a", "value_b");
```


RENAME a COLUMN in affiliations

```
CREATE TABLE affiliations (  
  firstname text,  
  lastname text,  
  university_shortname text,  
  function text,  
  organisation text  
);
```

```
ALTER TABLE table_name  
RENAME COLUMN old_name TO new_name;
```


DROP a COLUMN in affiliations

```
CREATE TABLE affiliations (  
  firstname text,  
  lastname text,  
  university_shortname text,  
  function text,  
  organization text  
);
```

```
ALTER TABLE table_name  
DROP COLUMN column_name;
```

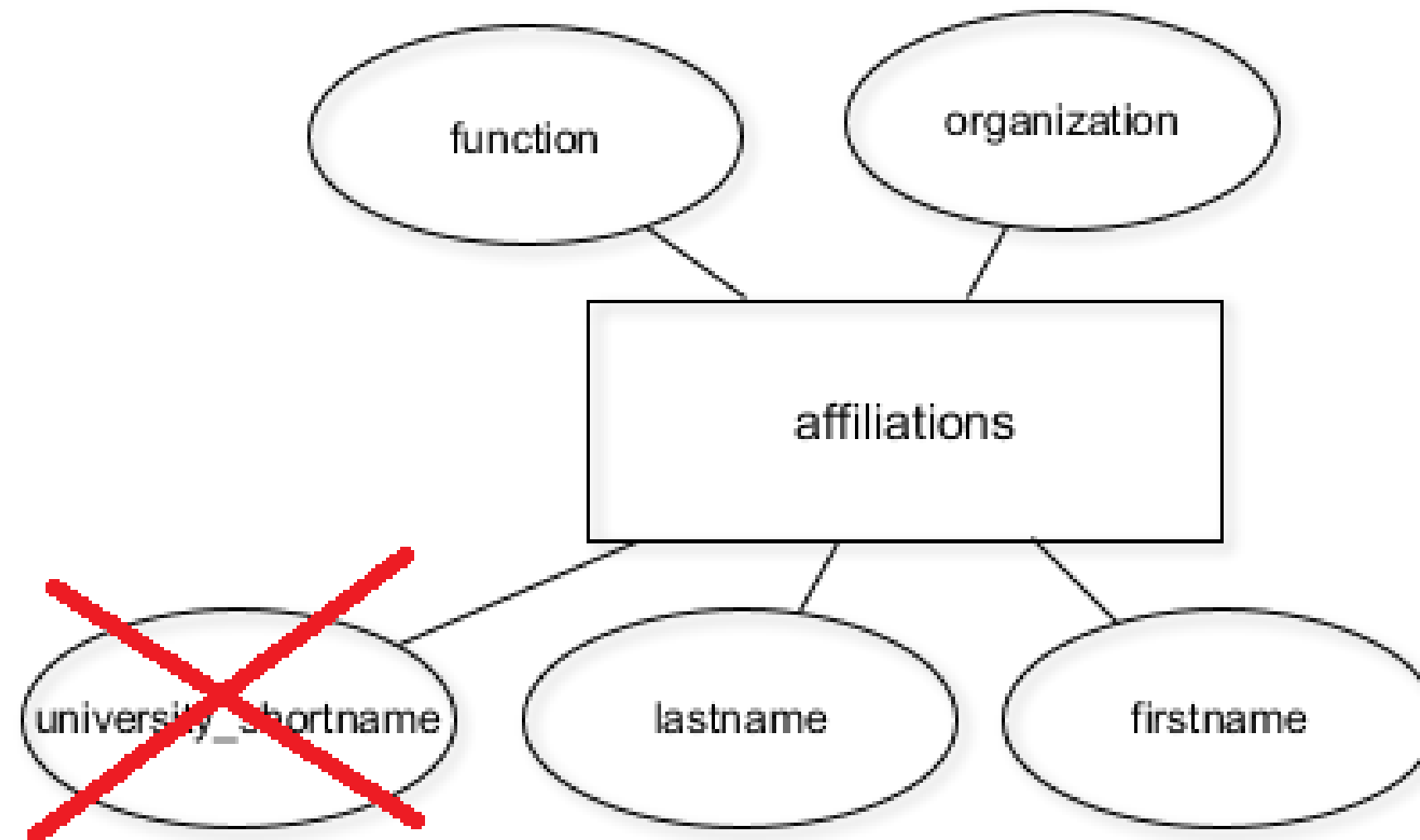
```
SELECT DISTINCT firstname, lastname,  
                university_shortcode  
FROM university_professors  
ORDER BY lastname;
```

```
-[ RECORD 1 ]-----+-----  
firstname      | Karl  
lastname       | Aberer  
university_shortcode | EPF  
-[ RECORD 2 ]-----+-----  
firstname      | Reza Shokrollah  
lastname       | Abhari  
university_shortcode | ETH  
-[ RECORD 3 ]-----+-----  
firstname      | Georges  
lastname       | Abou Jaoudé  
university_shortcode | EPF  
(truncated)  
  
(551 records)
```

```
SELECT DISTINCT firstname, lastname  
FROM university_professors  
ORDER BY lastname;
```

```
-[ RECORD 1 ]-----  
firstname | Karl  
lastname  | Aberer  
-[ RECORD 2 ]-----  
firstname | Reza Shokrollah  
lastname  | Abhari  
-[ RECORD 3 ]-----  
firstname | Georges  
lastname  | Abou Jaoudé  
(truncated)  
  
(551 records)
```

A professor is uniquely identified by firstname, lastname only



Let's get to work!

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